04. Word Codes

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Simple forward shift:

If every letter in a code shifts forward 2 places in the alphabet, what is the code for DOG?

Answer: D→F, O→Q, G→I (shifting each letter forward 2 places). Answer: **FQI**.

Example 2 — Multiple-choice, check every letter:

Using a shift of +3 (each letter moves 3 places forward), which is the correct code for FISH? A) ILVK B) ILUK C) IKVK D) HLVK E) ILVJ

Answer: Shifting each letter of FISH forward 3: F→I, I→L, S→V, H→K, giving ILVK. Answer: **A) ILVK**. Each wrong option changes just one letter, so check every letter carefully.

Example 3 — Select-all encoding check:

Using a shift of +1 (each letter moves 1 place forward), select ALL the words below that are correctly encoded. A) CAT → DBU B) DOG → EPG C) SUN → TVO D) BEE → CFF

Answer: CAT→DBU, SUN→TVO and BEE→CFF are all shifted correctly forward 1 place, so **A, C and D** are correct. DOG should become EPH (not EPG), so B is incorrect.

Example 4 — Decoding:

Using a shift of +2 (each letter moves 2 places forward when encoding), what word does the code ECV represent?

Answer: To decode, shift each letter back 2 places: E→C, C→A, V→T. Answer: **CAT**.

Example 5 — Shift that wraps around the alphabet:

Using a shift of +3, what is the code for LAZY? (Remember: after Z, the alphabet wraps back around to A.)

Answer: L→O, A→D, Z→C (Z wraps round past the end of the alphabet to C), Y→B (Y wraps round to B). Answer: **ODCB**.

Foundation

15 questions

1. If the code for FOX is GPY, what is the code for SKY?

2. If the code for FOX is GPY, what is the code for RUN?

3. If the code for FOX is GPY, what is the code for TOP?

4. If the code for FOX is GPY, what is the code for WIN?

5. If the code for FOX is GPY, what is the code for ICE?

6. If the code for FOX is GPY, what is the code for OWL?

7. If the code for FOX is GPY, what is the code for TEA?

8. If the code for FOX is GPY, what is the code for BAT?

9. If the code for FOX is GPY, what is the code for HAT?

10. If the code for FOX is GPY, what is the code for FAN?

11. Using the FOX → GPY code, what is the code for VAN? A) WBO B) UZN C) WAN D) VBP E) WBN

12. If the code for FOX is GPY, what is the code for ZOO?

13. If the code for FOX is GPY, what is the code for JOY?

14. Select ALL that apply: using the FOX → GPY code (shift +1), which of these are correctly encoded? A) PIE → QJF B) BEE → CFG C) ZOO → APP D) HAT → IBT E) TEA → UFB

15. If the code for FOX is GPY, what is the code for BEE?

Standard

20 questions

1. If the code for PLUM is SOXP, what is the code for SNOW?

2. If the code for PLUM is SOXP, what is the code for CLAM?

3. If the code for PLUM is SOXP, what is the code for DUSK?

4. If the code for PLUM is SOXP, what is the code for WAVE?

5. If the code for PLUM is SOXP, what is the code for GOLD?

6. Using the PLUM → SOXP code, what is the code for MINT? A) PLQW B) PLQX C) OLQW D) PKQW E) PLPW

7. If the code for PLUM is SOXP, what is the code for RIVER?

8. If the code for PLUM is SOXP, what is the code for STORM?

9. If the code for PLUM is SOXP, what is the code for FLAME?

10. If the code for PLUM is SOXP, what is the code for GLOBE?

11. If the code for PLUM is SOXP, what is the code for PEARL?

12. If the code for PLUM is SOXP, what is the code for BRICK?

13. If the code for PLUM is SOXP, what is the code for CRANE?

14. If the code for PLUM is SOXP, what is the code for FROST?

15. If the code for PLUM is SOXP, what is the code for QUEEN?

16. Select ALL the words below that are correctly encoded using the PLUM → SOXP code (shift +3). A) SHARK → VKDUN B) WOLF → ZROI C) MOUSE → PRXVG D) CRANE → FUDQF E) GLOBE → JOREH

17. Using the PLUM → SOXP code (shift +3), what word does YRLFH represent?

18. Using the PLUM → SOXP code (shift +3), what word does ZLQWHU represent? A) WINTER B) WINDOW C) WONDER D) WRITER E) WAITER

19. Select ALL the words below that correctly decode the codes given, using the PLUM → SOXP code (shift +3). A) IODPH = FLAME B) SKDUN = SHARK C) EULFN = BRICK D) YRLFH = VOICE E) MRXVH = MOUSE

20. Using the PLUM → SOXP code (shift +3), what word does JOREH represent?

Hard

5 questions

1. Using the PRIDE → TVMHI code (shift +4), what is the code for BRAVE?

2. Using the PRIDE → TVMHI code (shift +4), what word does LSYVW represent? A) HOURS B) HOUSE C) HORNS D) HOVER E) HONEY

3. Select ALL the words below that are correctly encoded using the PRIDE → TVMHI code (shift +4). A) CLOUD → GPSXH B) STORM → WXSVQ C) KNIGHT → OVKIRX D) SPARK → WTEVO E) FLAME → JPEQI

4. Using the PRIDE → TVMHI code (shift +4), what word does GEWXPI represent?

5. The code for FROZEN is JVSDIR (the same +4 shift as PRIDE → TVMHI, but notice how Z wraps around to become D). What is the code for ZEBRA, using this same shift?

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